

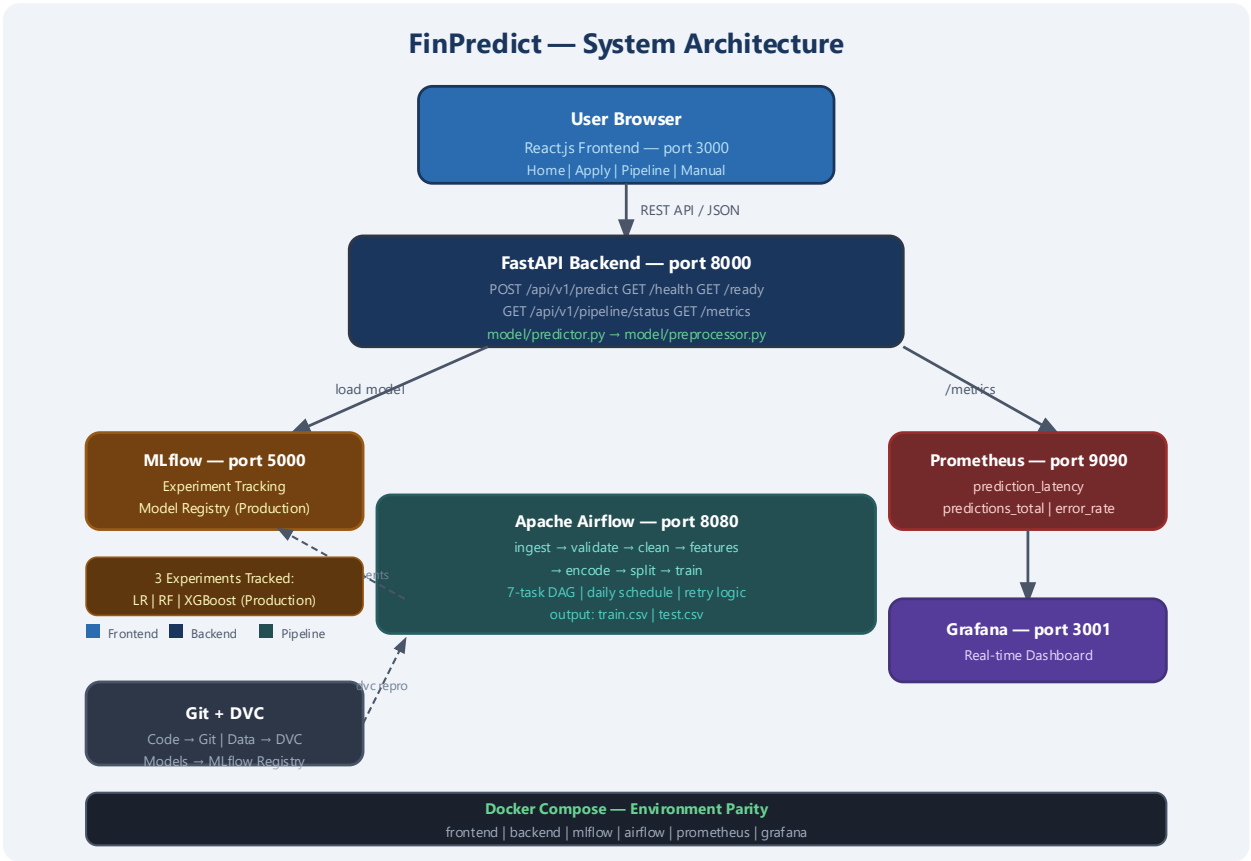
FinPredict

Architecture Diagram & Block Explanation

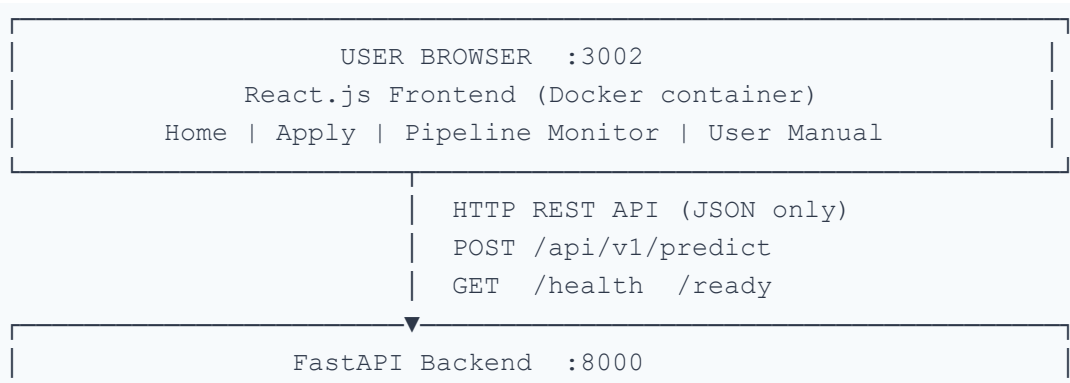
System Design Document | Version 1.0 | April 2026

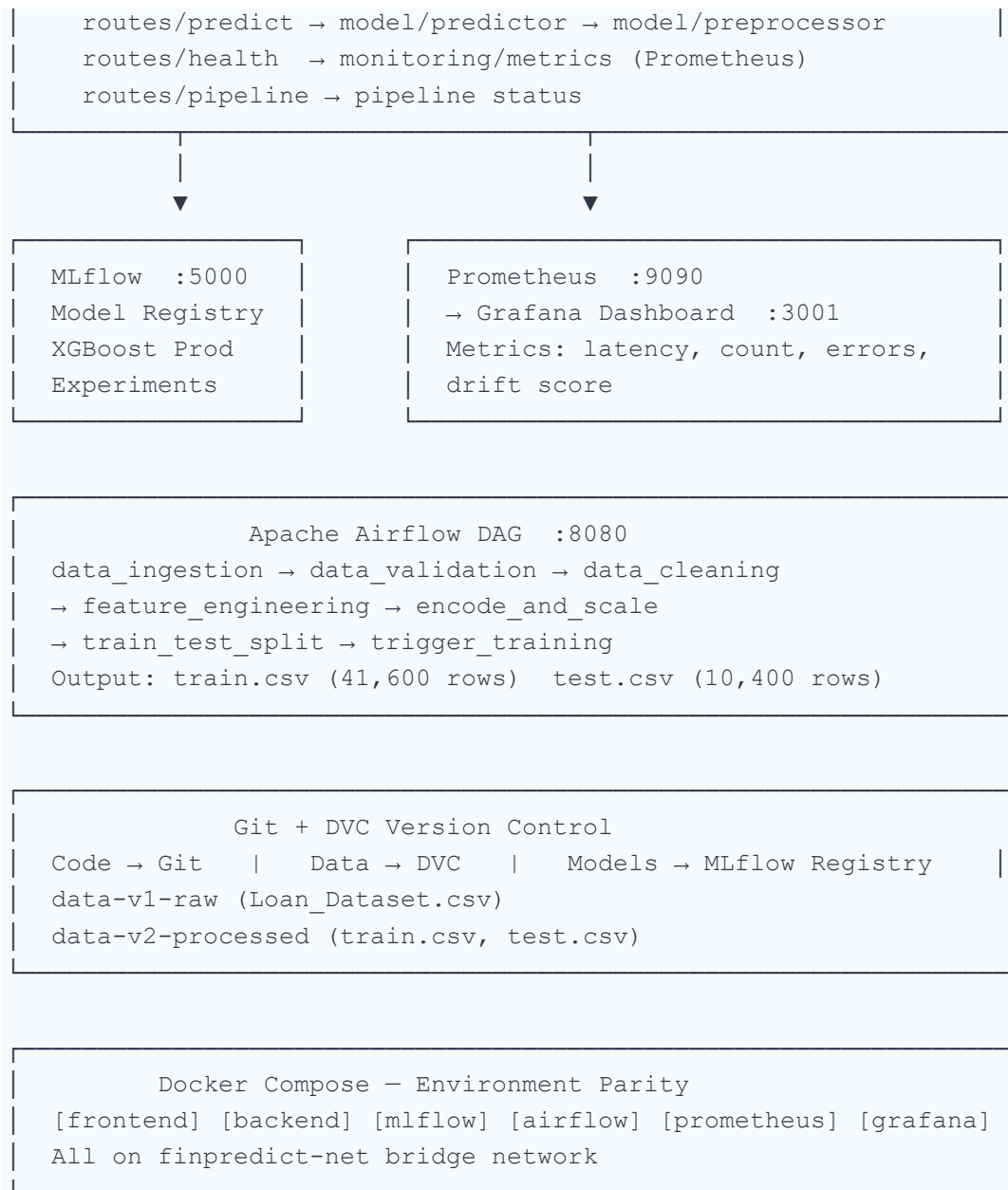
1. Architecture Overview

FinPredict follows a microservices architecture where each component runs as an independent Docker container. All services share a private Docker bridge network (finpredict-net) and communicate via HTTP. The frontend communicates with the backend exclusively through REST API calls, enforcing strict loose coupling as required by the evaluation guidelines.



2. Architecture Diagram — ASCII Representation



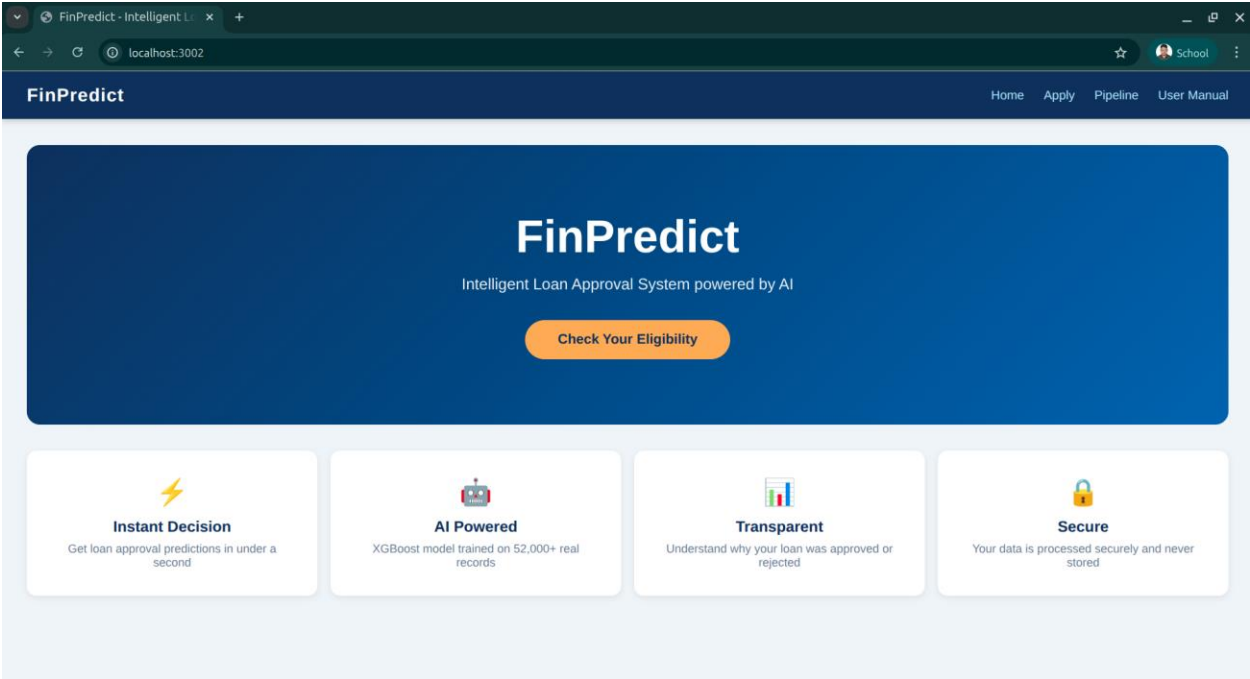


3. Component Block Explanations

3.1 Frontend (React.js — Port 3002)

The frontend is a React.js single-page application serving as the user interface for loan applications. It has zero direct access to the model, database, or preprocessing logic.

Property	Detail
Technology	React.js 18, React Router, Axios
Port	3002 (mapped from 3000 inside container)
Docker Image	Built from frontend/Dockerfile using node:18-alpine
Communication	REST API calls only to http://localhost:8000
Pages	Home, Apply/Predict, Pipeline Monitor, User Manual
Key Dependency	REACT_APP_API_URL environment variable (configurable)

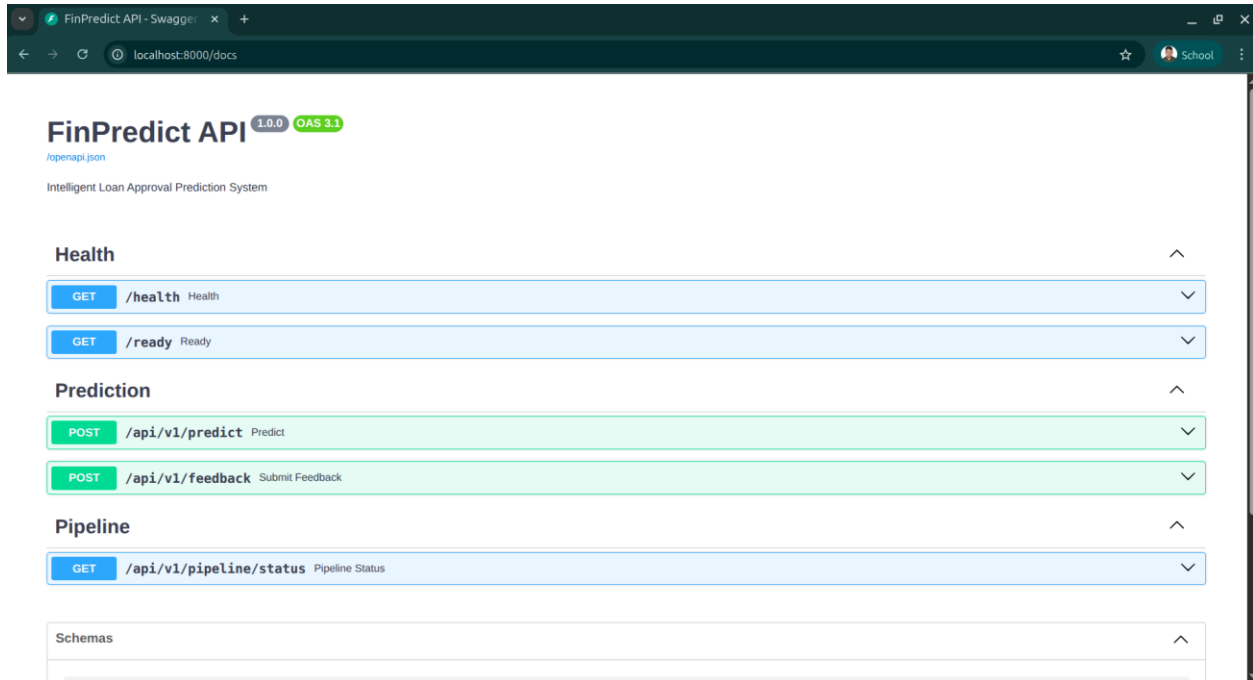


3.2 Backend (FastAPI — Port 8000)

The backend is the central inference engine. It loads the XGBoost model from MLflow at startup and serves all REST API requests from the frontend.

Property	Detail
Technology	Python 3.10, FastAPI 0.104, Uvicorn
Port	8000

Docker Image	Built from backend/Dockerfile using python:3.10-slim
Model Loading	Loaded from MLflow at startup: models:/FinPredict_XGBoost/Production
Key Endpoints	/predict, /health, /ready, /feedback, /pipeline/status, /metrics
Monitoring	Prometheus metrics exposed at /metrics
Logging	Structured logging to backend/logs/app.log



3.3 MLflow Model Registry (Port 5000)

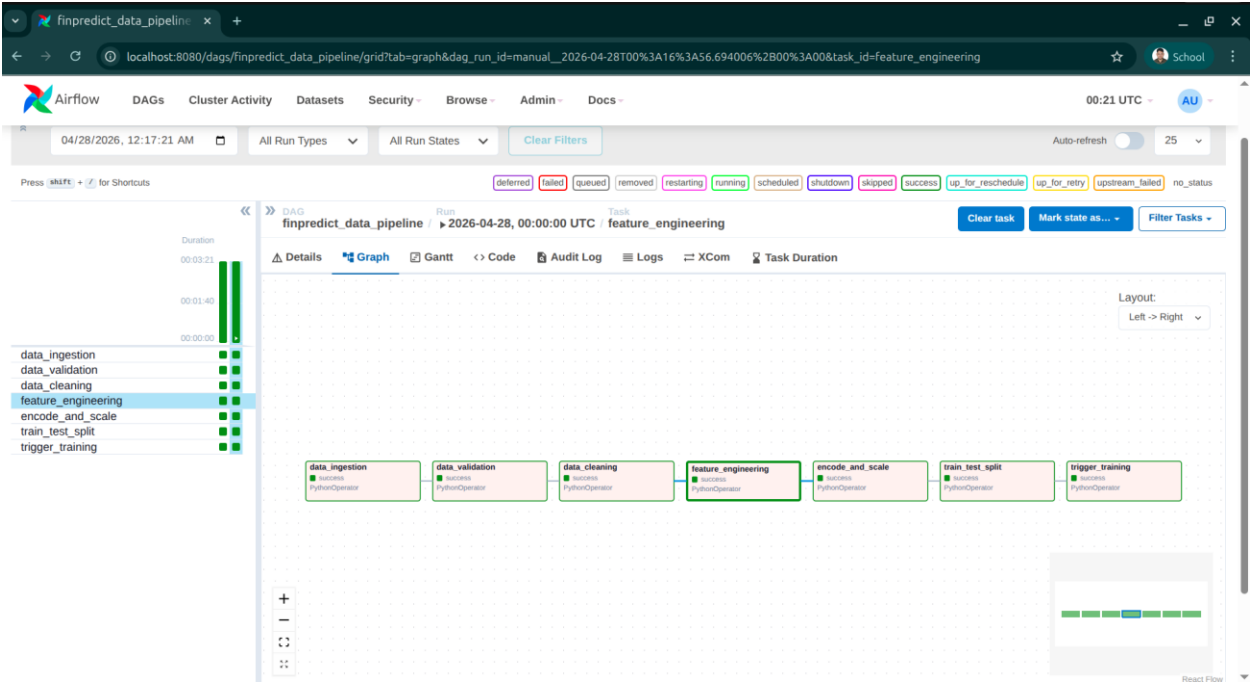
MLflow serves as the single source of truth for all model versions and experiments. The backend loads the Production-stage model from this registry.

Property	Detail
Technology	MLflow 2.19.0
Port	5000
Docker Image	ghcr.io/mlflow/mlflow:v2.19.0
Storage	SQLite backend + /mlflow/artifacts volume
Experiments	FinPredict_LoanApproval (3 runs: LR, RF, XGBoost)
Registry	FinPredict_XGBoost — Version 2 in Production stage
Extra Tracking	Feature importance, classification report, baseline stats as artifacts

3.4 Apache Airflow (Port 8080)

Airflow orchestrates the 7-task data engineering DAG that runs daily and can be triggered manually. It manages retries, scheduling, and failure tracking.

Property	Detail
Technology	Apache Airflow 2.7.0 (SequentialExecutor + SQLite)
Port	8080
Docker Image	Built from airflow/Dockerfile (custom with sklearn, pandas, xgboost)
DAG	finpredict_data_pipeline — 7 PythonOperator tasks
Schedule	@daily with manual trigger support
Retry	1 retry per task with 5-minute delay
Mounted Volumes	./mlops/data_pipeline → /opt/airflow/dags, ./data → /data



3.5 Prometheus (Port 9090)

Prometheus scrapes the backend /metrics endpoint every 15 seconds and evaluates 4 alert rules for error rate, latency, no predictions, and data drift.

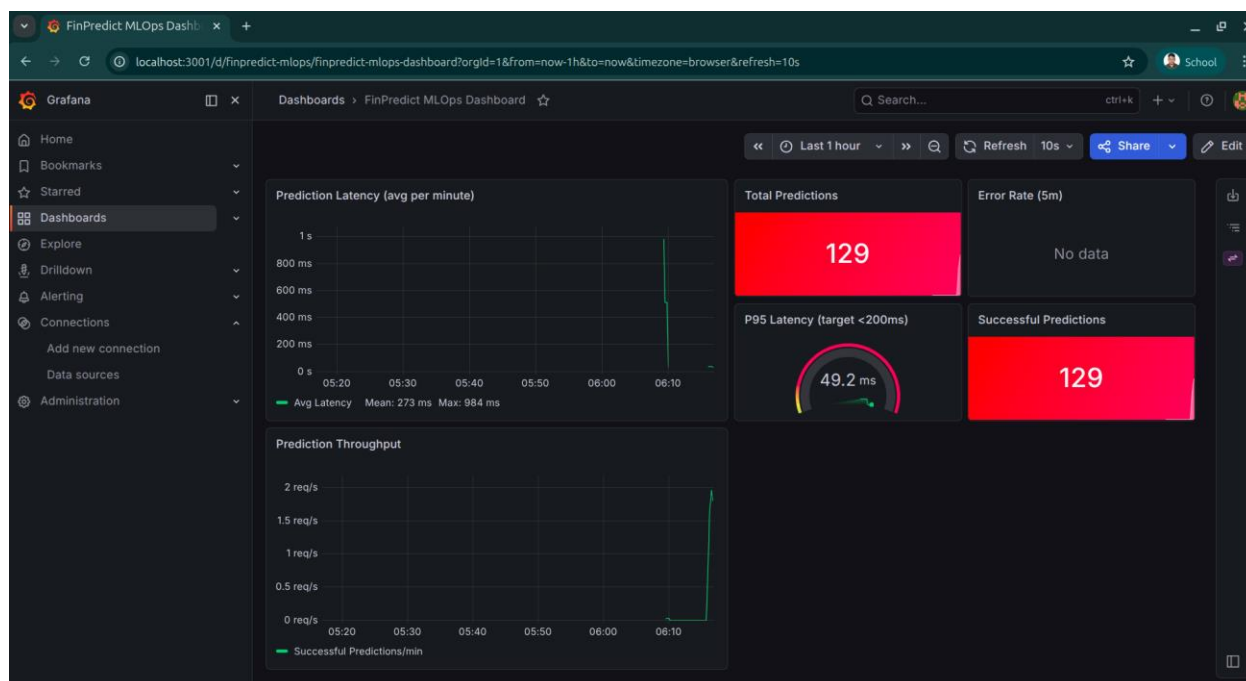
Property	Detail
Technology	Prometheus latest
Port	9090
Scrape Interval	15 seconds

Scrape Target	backend:8000/metrics
Alert Rules	4 rules in monitoring/alert_rules.yml
Alertmanager	Email notifications to Gmail when alerts fire

3.6 Grafana (Port 3001)

Grafana provides a real-time dashboard with 6 panels, auto-provisioned via configuration files — no manual setup required.

Property	Detail
Technology	Grafana latest
Port	3001
Datasource	Auto-provisioned Prometheus at http://prometheus:9090
Dashboard	Auto-loaded from monitoring/grafana/dashboards/finpredict_dashboard.json
Refresh Rate	10 seconds
Panels	Latency, Total Predictions, Error Rate, Throughput, P95 Latency, Success Count



4. Component Interaction Summary

From	To	Protocol	Purpose
Browser	Frontend Container	HTTP	Load React application
Frontend	Backend :8000	HTTP REST / JSON	Send prediction requests
Backend	MLflow :5000	HTTP	Load Production model at startup
Backend	Prometheus :9090	HTTP (scrape)	Expose metrics at /metrics
Prometheus	Grafana :3001	HTTP	Supply metrics for dashboard
Prometheus	Alertmanager	HTTP	Fire alerts when rules trigger
Alertmanager	Gmail	SMTP	Send email notifications
Airflow	Data files	File I/O	Read/write CSV pipeline data
Airflow	MLflow :5000	HTTP	Log experiments and register model
DVC	Git	File tracking	Version data and pipeline stages
Local machine	MLflow :5000	HTTP	Run experiments via Python scripts